

CANDIDATE NAME: _____

INDEX NUMBER _____

CG _____



SERANGOON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2018

H1 BIOLOGY

Paper 2 Structured and Free-response Questions

8876/02

Thursday
13 September 2018

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, index number and CG in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE	
Paper 1 (MCQ)	/30
Paper 2	
Section A	
1	/6
2	/8
3	/11
4	/9
5	/11
Section B	
6 or 7	/15
P2 Total	/60
Total	/90
TOTAL (100%)	

This question paper consists of **19** printed pages including this cover page.

Section A

Answer **all** the questions in this section.

1. Fig. 1.1 is an electron micrograph of part of an animal cell.

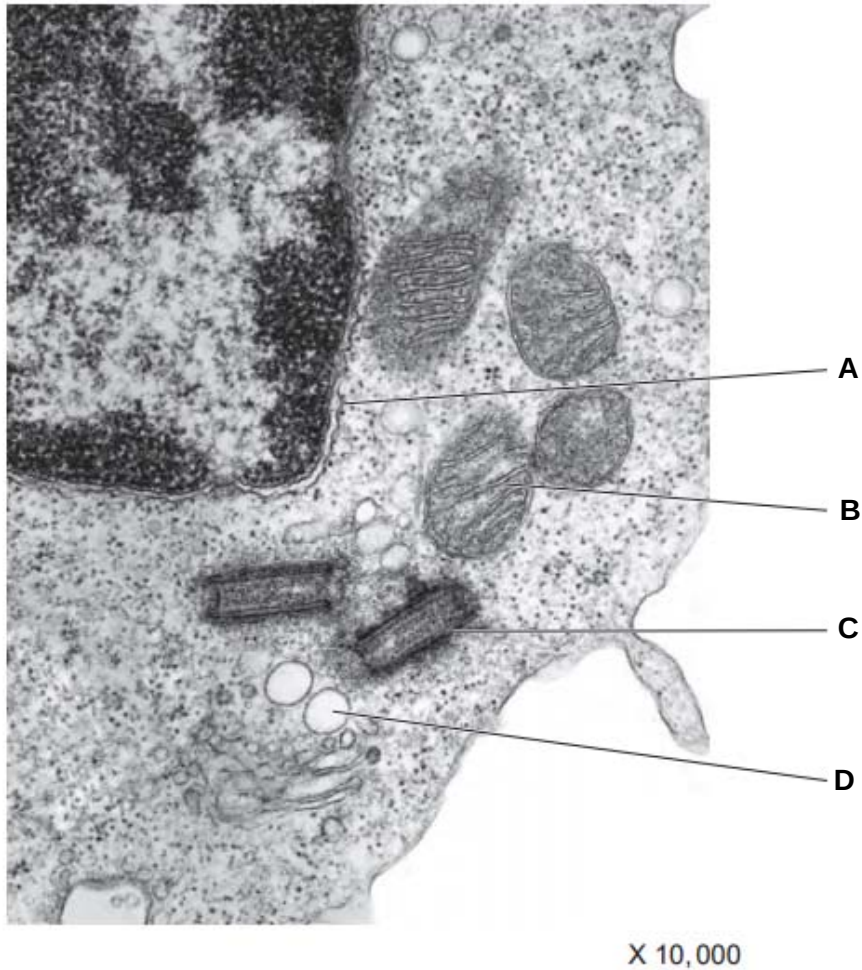


Fig. 1.1

- (a) Name the membranes labelled **A** and **B**.

A

B [2]

- (b) Describe the role of structure **C** in cell division.

.....
.....
.....
..... [2]

(c) Explain how it possible for the pH in structure **D** to be kept constant.

.....

.....

.....

.....[2]

[Total: 6]

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2. Fig. 2.1 is an electron micrograph of a cancer cell in the midst of mitosis.

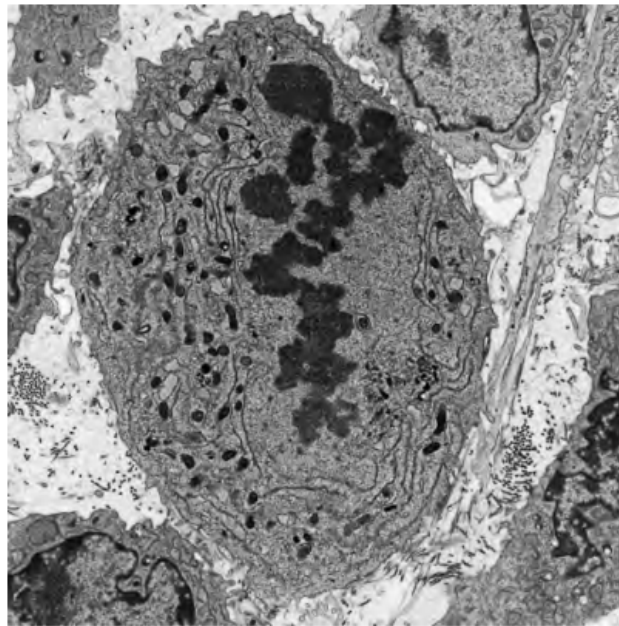


Fig. 2.1

- (a)** Identify the stage of mitosis, giving a reason for your answer.

stage

reason

.....
[2]

People who have smoked cigarettes for many years are at risk of developing lung cancer.

- (b)** Explain why long-term smoking increases the risk of cancer in smokers.

.....
.....
.....
.....[2]

Fig. 2.2 shows the change in the percentage of smokers in the male population of the UK between 1950 and 2005. Fig. 2.3 shows the change in mortality rate in the UK in men aged 75 to 84 between 1950 and 2005.

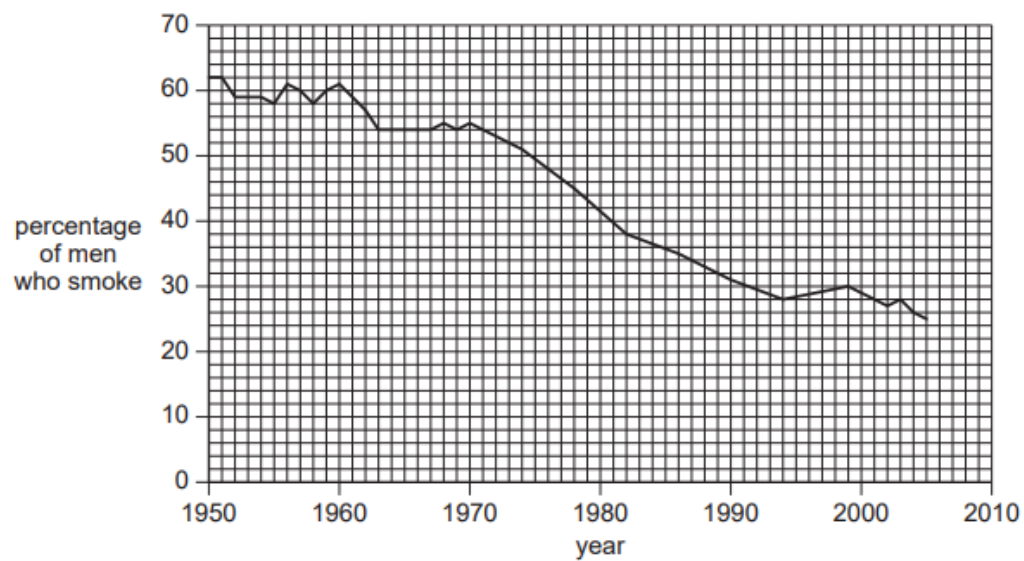


Fig. 2.2

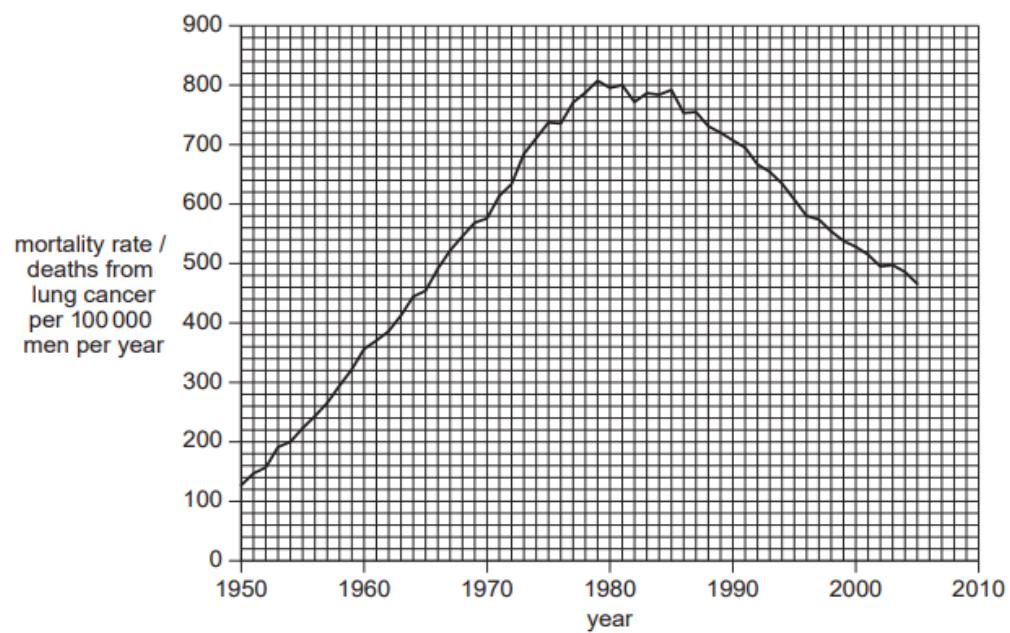


Fig. 2.3

(c) With reference to Figs. 2.2 and 2.3, discuss the observations made between 1950 and 2005.

.....

.....

.....

.....

.....

.....

.....

.....[4]

[Total: 8]

3. DNA and RNA are some of the key molecules involved in the synthesis of proteins.

(a) Comment on how DNA and tRNA are different, in terms of the bases present.

.....
.....
.....
.....[2]

(b) Calculate the minimum number of DNA nucleotides required to code for a polypeptide with 238 amino acids.

- Show your working in the space below.
- State the **two** properties of the genetic code that are considered in deriving your answer.

.....
.....
.....
.....[3]

(c) Describe the role played by tRNA in polypeptide synthesis.

.....
.....
.....
.....
.....
.....[3]

- (d) Regulation can occur following the synthesis of proteins. An example of such regulation is shown in Fig. 3.1, where inactive enzyme precursor pepsinogen is modified post-translationally to form active pepsin.

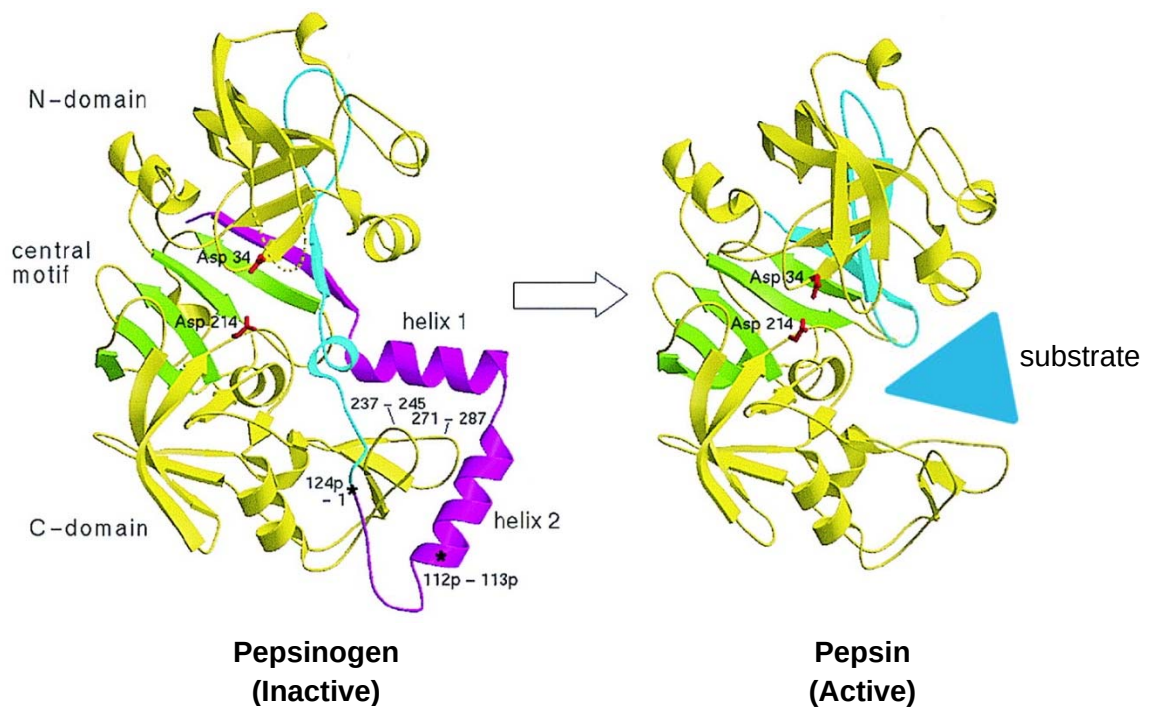


Fig. 3.1

- (i) With reference to Fig. 3.1, describe the post-translational change observed in the enzyme and explain why it results in its activation.

.....

[2]

- (ii) Suggest why pepsin is synthesised as an inactive precursor pepsinogen.

.....
[1]

[Total: 11]

4. The fruit fly, *Drosophila melanogaster*, has eyes, a striped abdomen and wings longer than its abdomen. This is called a 'wild-type' fly.

Mutation has resulted in many variations of these features. Table 4.1 shows diagrams of a wild-type fly and three other flies, each of which shows one **recessive mutation**.

Table 4.1

				
eyes	present	present	absent	present
abdomen	striped	black	striped	striped
wing description	long	long	long	short

- (a) Explain what is meant by a "recessive mutation".

.....

[2]

- (b) Using appropriate symbols, illustrate a cross between a fly without eyes and long wings and one with eyes and short wings that would result in **four** different phenotypes being observed in the offspring.

[4]

- (c) The class of mutations giving rise to the various mutant phenotypes in Table 4.1 are termed as **gene mutations**.

Explain how these mutations are different from chromosomal aberrations.

.....

.....

.....

.....

.....

.....[3]

[Total: 9]

5. ATP and NADP both play important roles in photosynthesis. Fig. 5.1 represents the molecular structures of ATP and NADP.

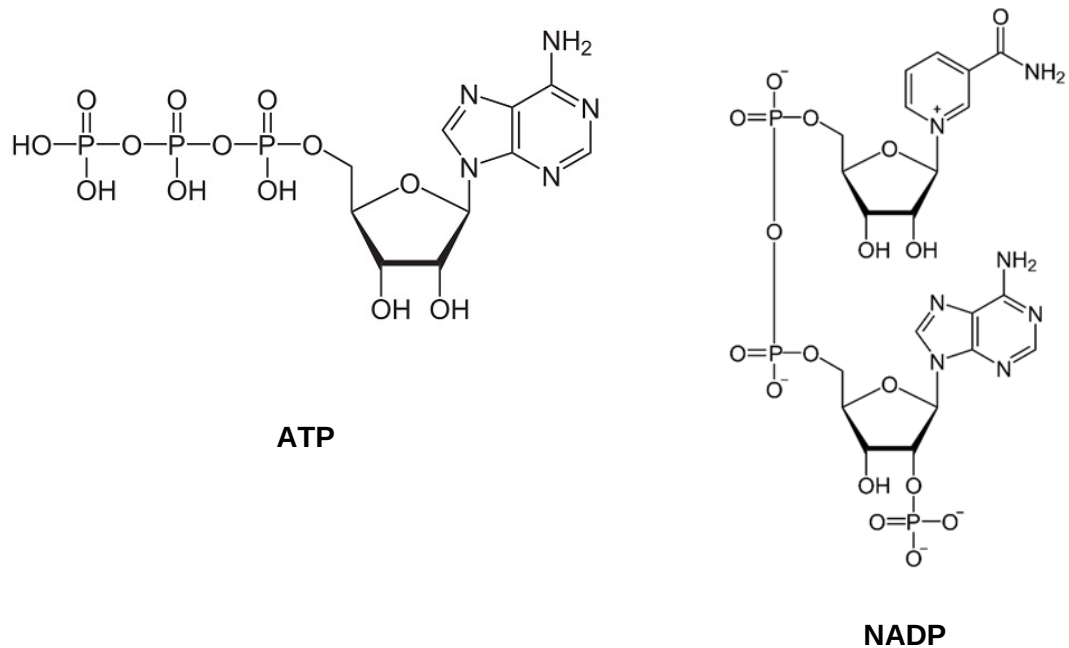


Fig. 5.1

- (a)** Using Fig. 5.1, compare the structures of ATP and NADP.

.....

.....

.....

.....

.....

.....[3]

- (b)** Outline the roles of NADP in a cell.

.....

.....

.....

.....[2]

- (c) State the names of the processes in which ATP is synthesised during photosynthesis and respiration respectively.

photosynthesis

.....

respiration

.....

[2]

- (d) ATP serves as a source of energy for several metabolic processes in both photosynthesis and respiration.

Name one process in **respiration** that requires ATP as an energy source.

.....[1]

- (e) The first substrate used in respiration is glucose. In a situation of excess glucose, some of these glucose is stored as fats instead of carbohydrates.

Explain why animals prefer to store lipid instead of carbohydrates.

.....

.....

.....

.....

.....

.....[3]

[Total: 11]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)** and **(b)**, as indicated in the question.

6. **(a)** Explain how climate change affects the spread of dengue beyond the tropics. [5]

- (b)** Discuss how variation is generated in a population, and explain why it is important in natural selection. [10]

[Total: 15]

7. **(a)** Explain how man's food choices can lead to climate change impacts. [5]

- (b)** Explain what is meant by the fluid mosaic model, and discuss the importance of membrane fluidity in biological processes. [10]

[Total: 15]

[illegible]

[illegible]

[illegible]

This image shows a full page of white paper with horizontal dotted lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

[Total: 15]